

3. Forensic scientists analyse substances found at crime scenes. They routinely identify powder samples.

Students are using the same techniques to analyse an unknown powder.

- (a) The students dissolve the unknown powder in water and add sodium hydroxide solution (NaOH) to test for metal cations. They then add excess NaOH solution to see if the precipitate dissolves or not.

The table shows results for a selection of cations.

Cation	Colour of precipitate when a little NaOH solution is added	Precipitate dissolves when excess NaOH solution is added (✓ or ×)
Ca^{2+}	white	×
Zn^{2+}	white	✓
Pb^{2+}	white	✓
Mg^{2+}	white	×
Cu^{2+}	blue	×
Fe^{2+}	green	×

They noticed that a white precipitate was produced which did not dissolve in excess NaOH solution.

- (i) State the symbols for the **two** metal cations that could be present. [1]

.....

- (ii) The table below shows the flame tests for a selection of metal cations.

Metal cation	Flame colour
Ca^{2+}	brick-red
Zn^{2+}	greyish-white
Pb^{2+}	greyish-white
Mg^{2+}	no colour
Cu^{2+}	blue-green
Fe^{2+}	gold

- Describe how a flame test could be used to confirm which cation is present. [2]

.....

- (b) They use another sample of the powder and perform the following tests to identify the anions that may be present.

Anion	Test	Observation
bromide (Br^-)	add aqueous silver nitrate	cream precipitate
carbonate (CO_3^{2-})	add hydrochloric acid	bubbles of gas
chloride (Cl^-)	add aqueous silver nitrate	white precipitate
iodide (I^-)	add aqueous silver nitrate	yellow precipitate
sulfate (SO_4^{2-})	add aqueous barium chloride	white precipitate

The observations from the students' tests are:

- bubbles of gas are given off when hydrochloric acid is added
- a white precipitate forms when aqueous silver nitrate is added

- Use the information above to determine the **two** anions present in the powder. [2]

.....

- (c) Name **one** compound that could be present in the powder. [1]

.....